



EC2.103

Analog Electronic Circuits

Quadrature Down Converter

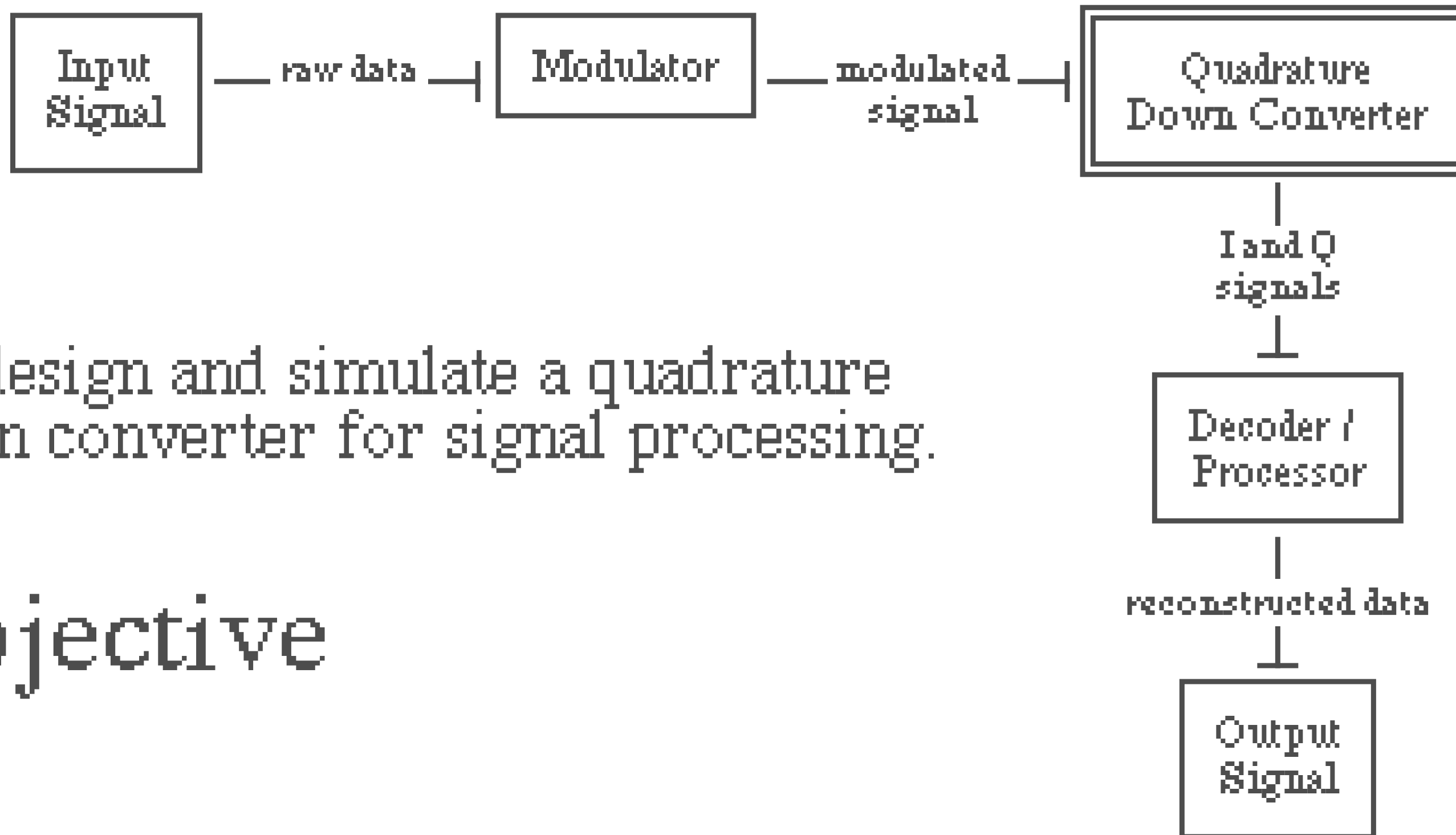
Project mid-evaluation

small enemy spider

Harry Jain 2024112001

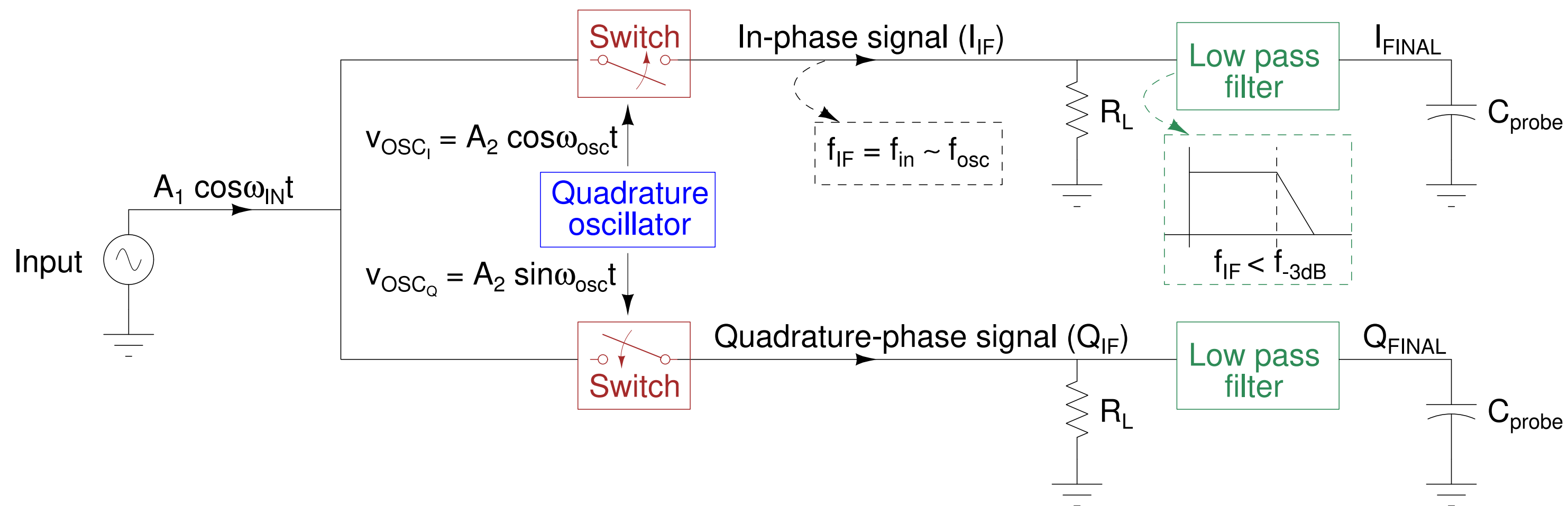
Kshitij Patkar 2024112005

Sanshray Vinesh 2024102071



To design and simulate a quadrature down converter for signal processing.

Objective



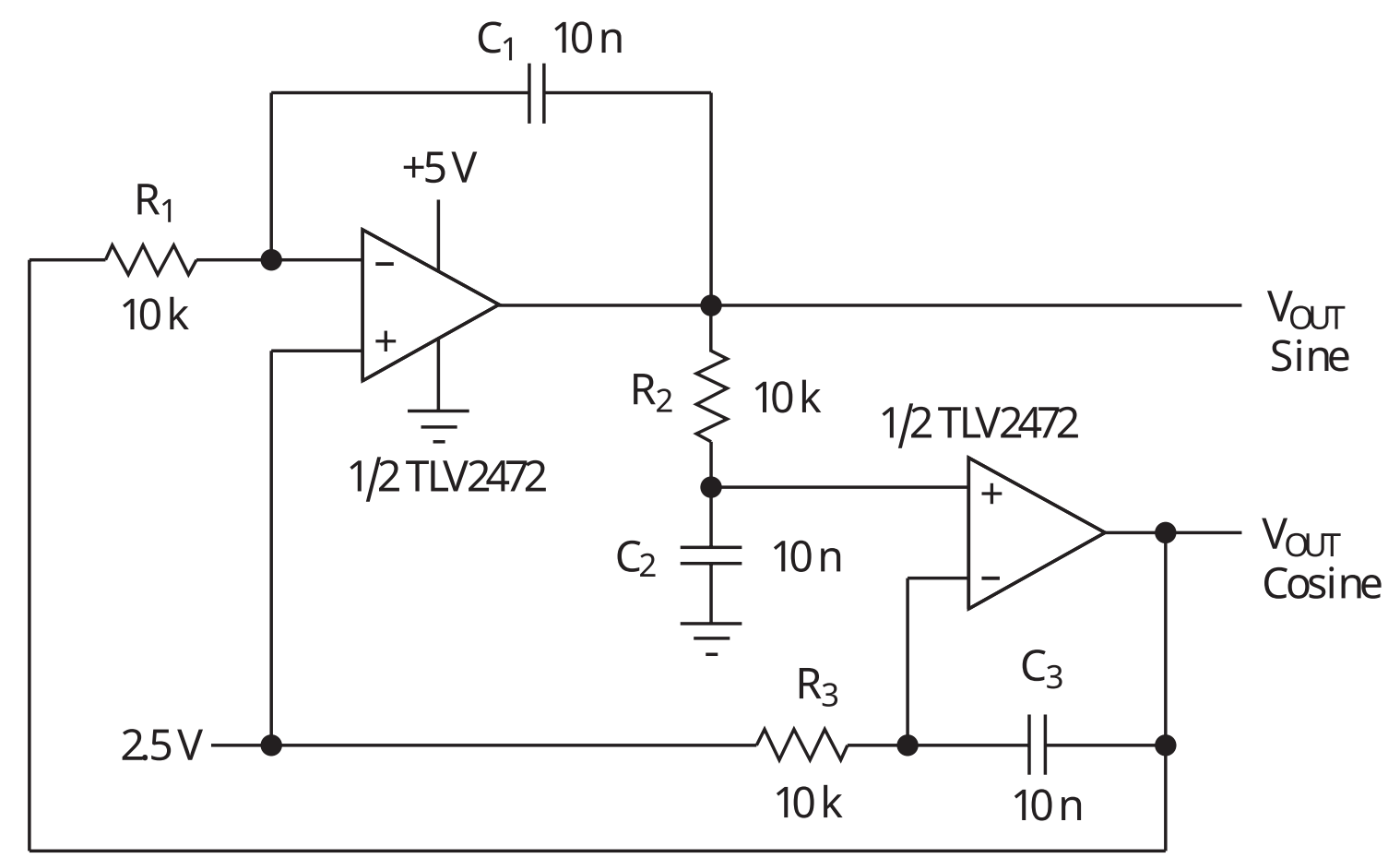
Block Diagram

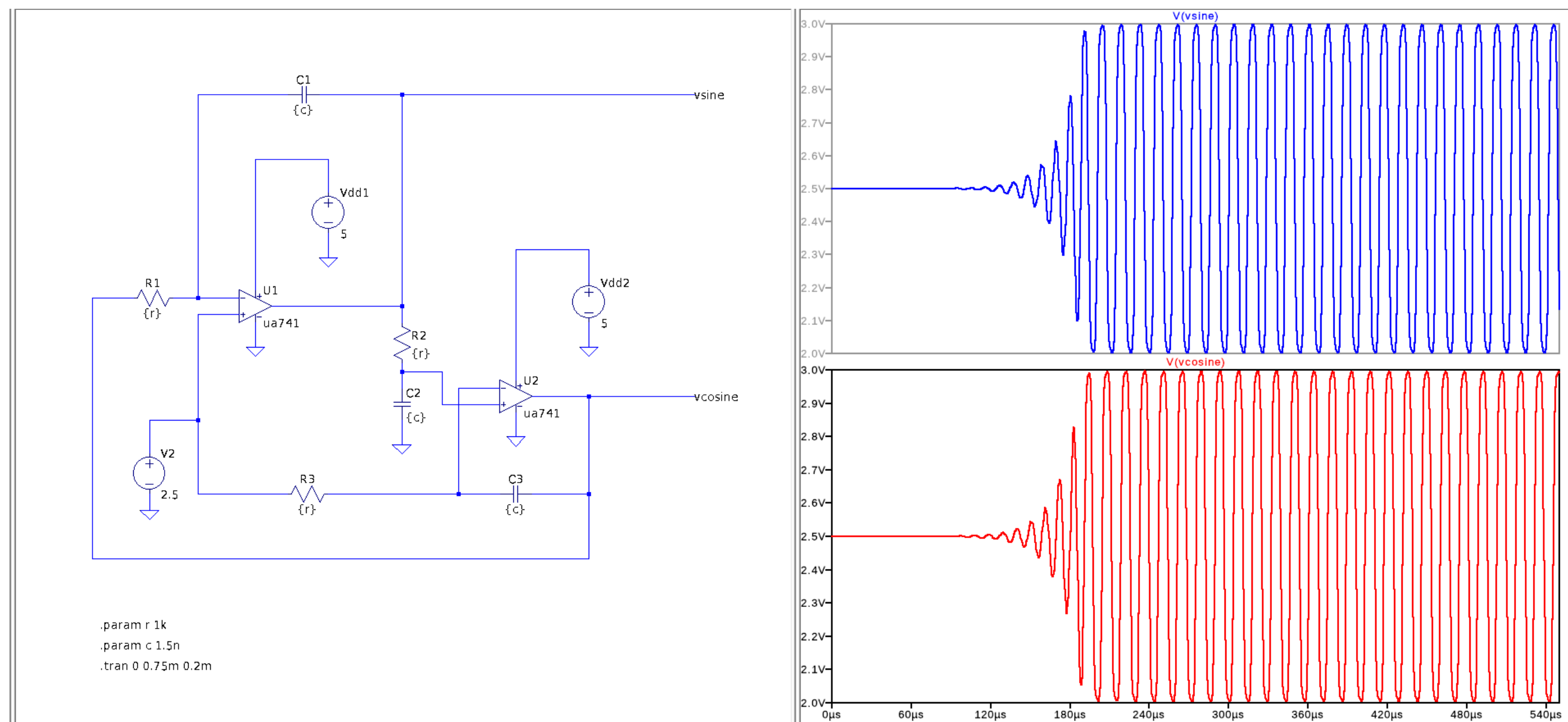


Quadrature Oscillator

Generates two sinusoidal signals
with a phase difference of 90 deg.

Local Oscillator



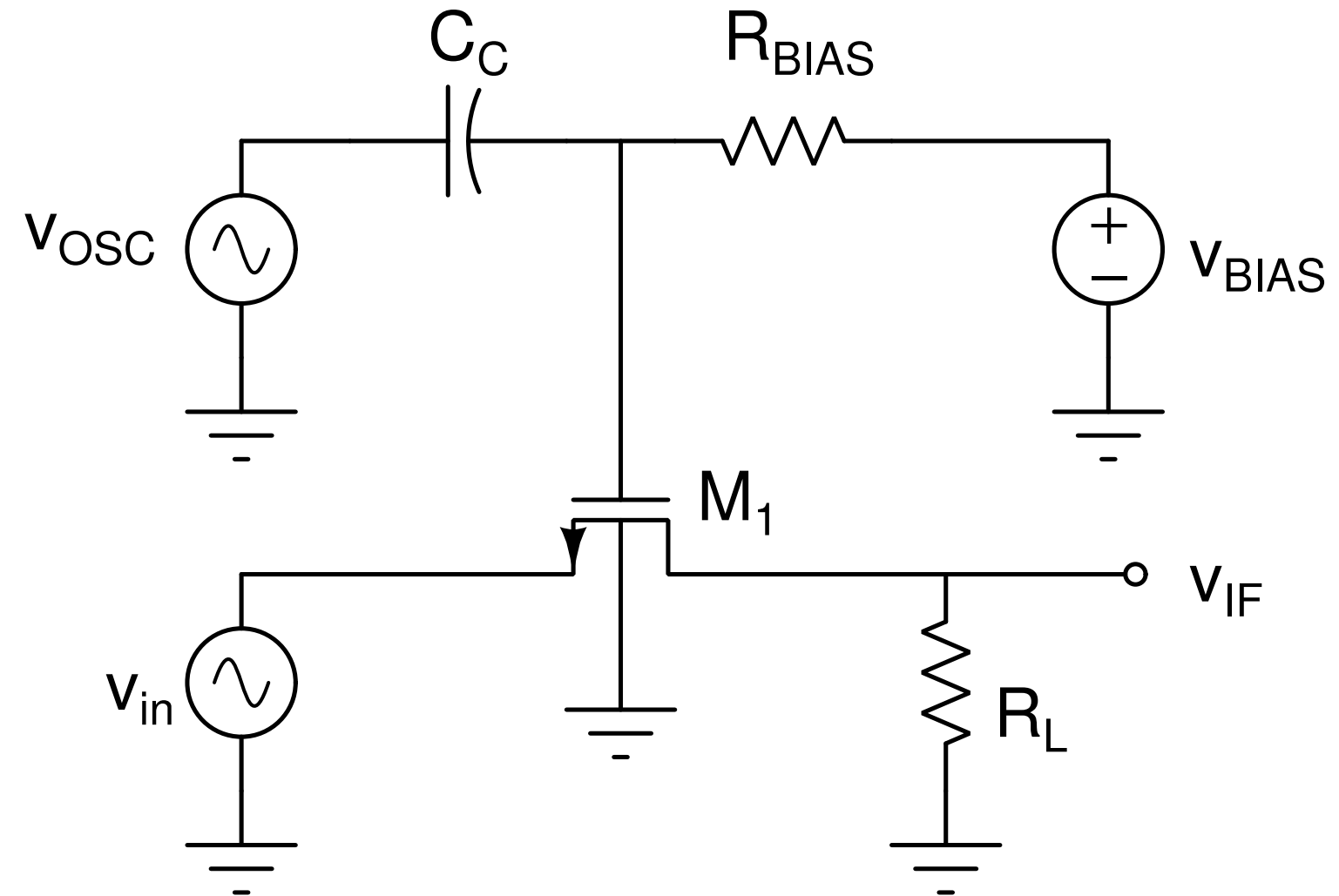


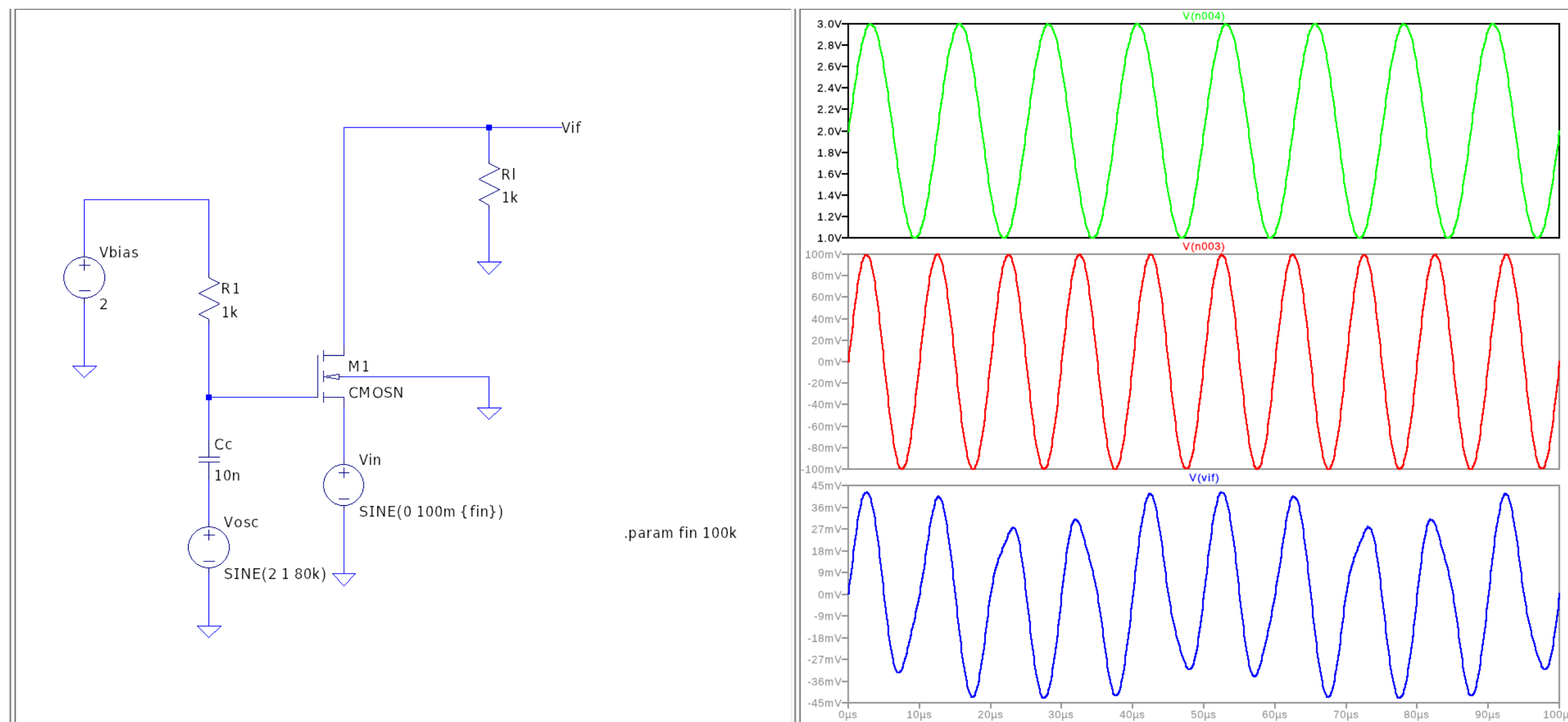
Quadrature oscillator circuit with two sinusoidal outputs

Multiplier

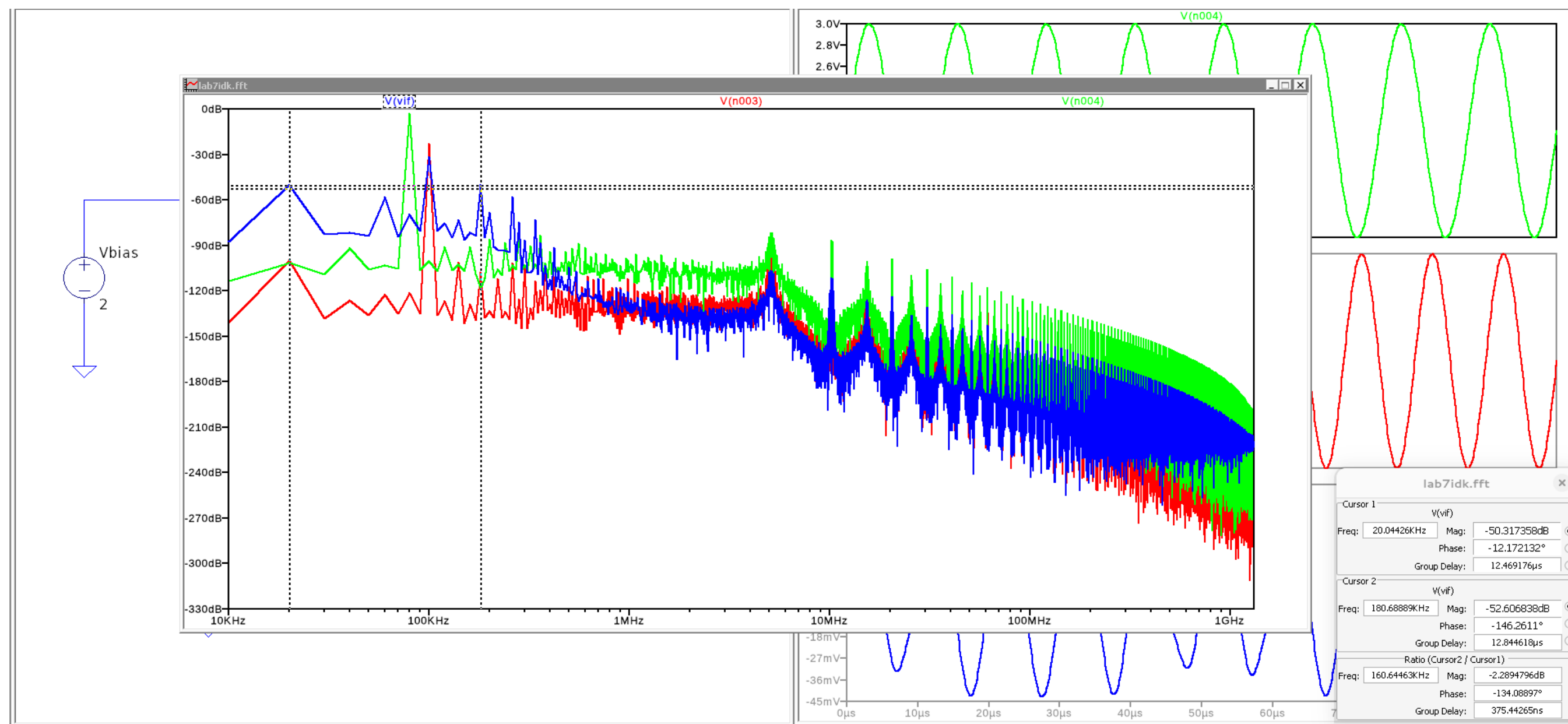
Multiplies the input signal with the biased oscillator signals

Mixer (Switcher)

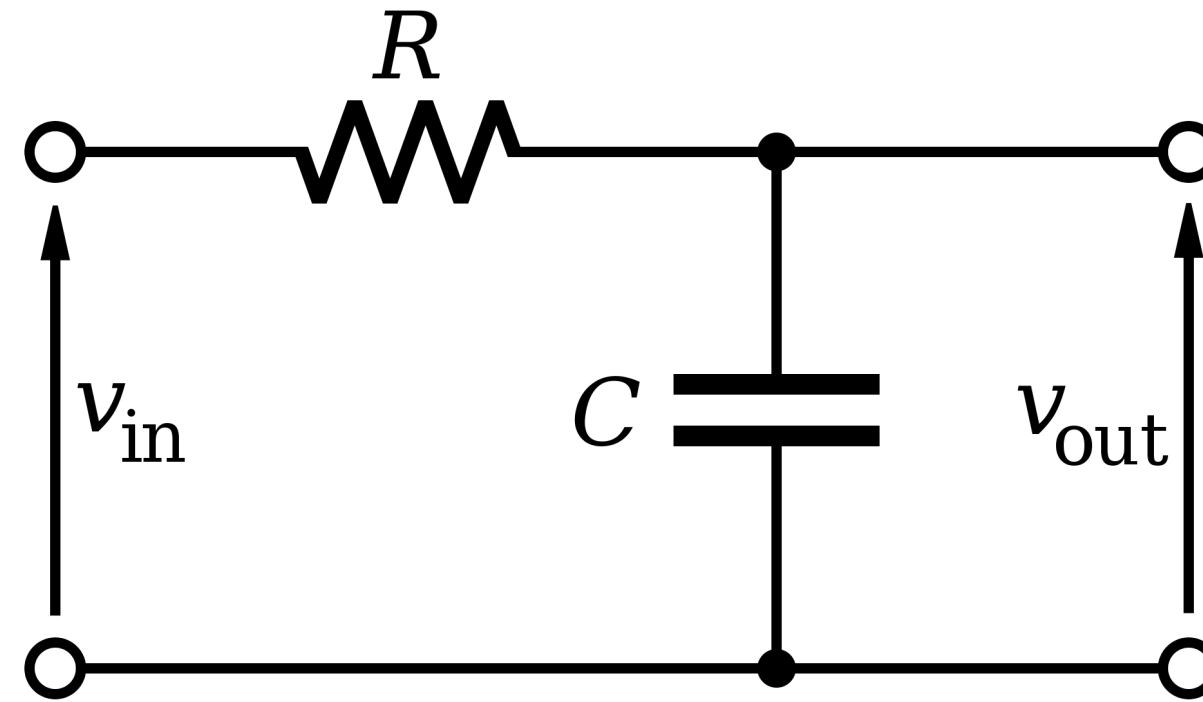




Multiplied output from an input and an oscillator signal

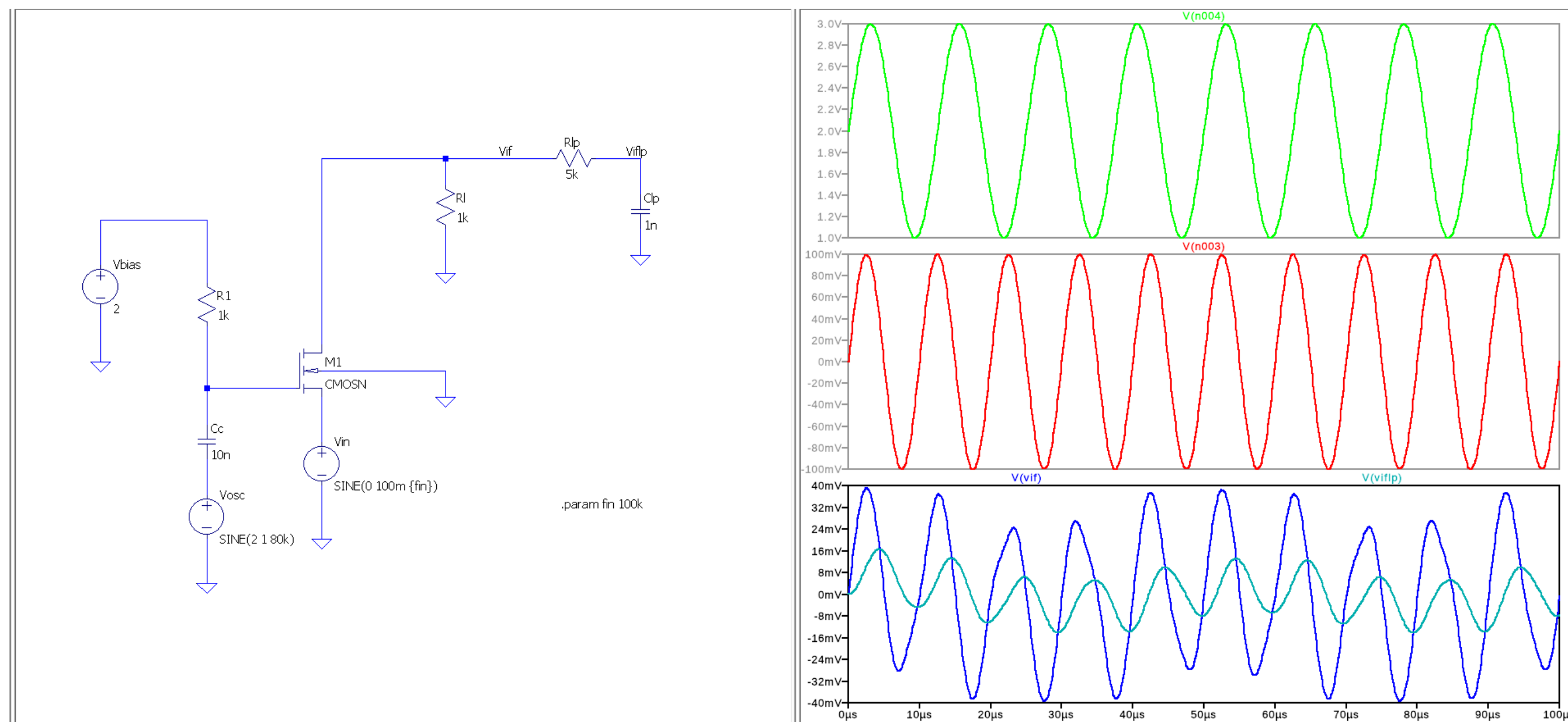


Frequency analysis of relevant signals, observe the 180 kOhm sum and 20 kOhm difference frequencies in the plot

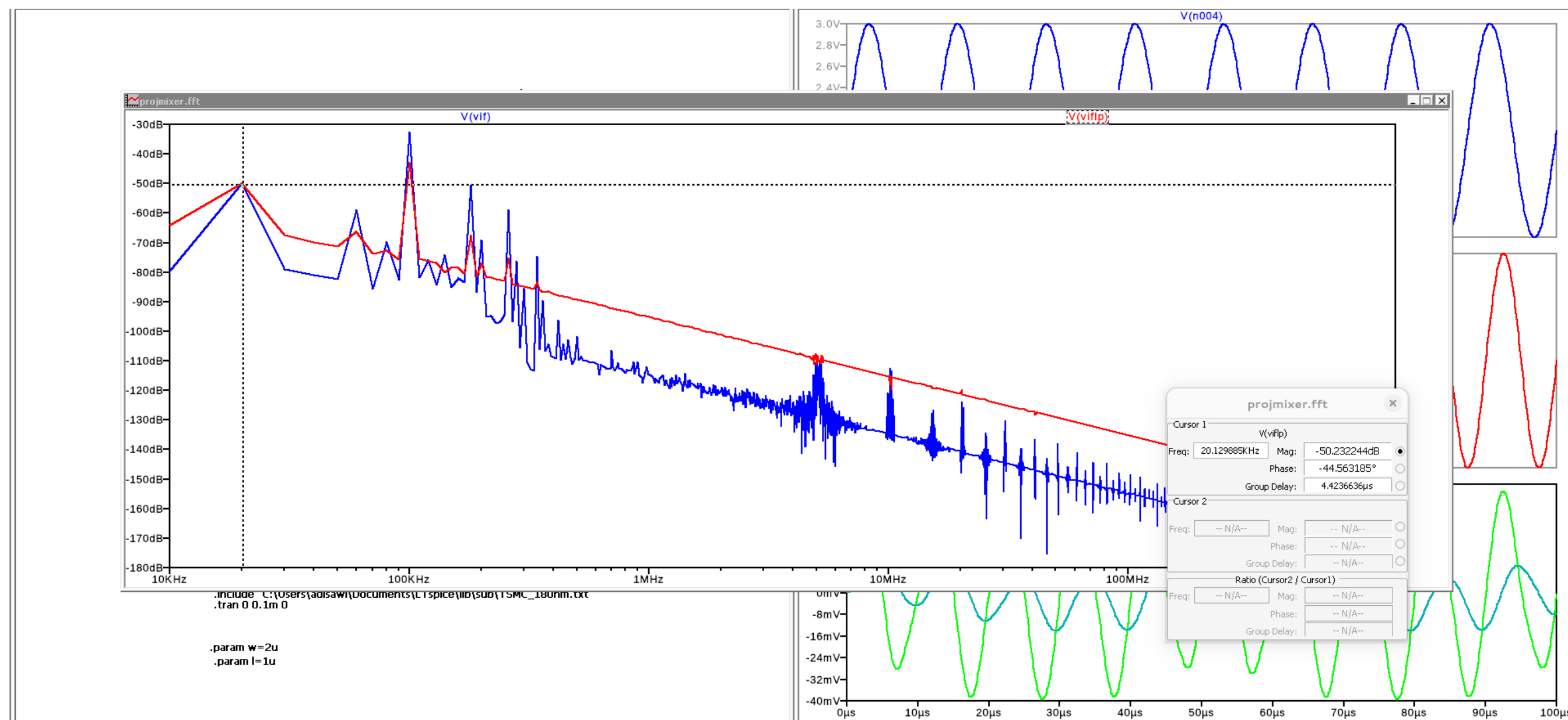


Stops the junk higher
frequencies from passing

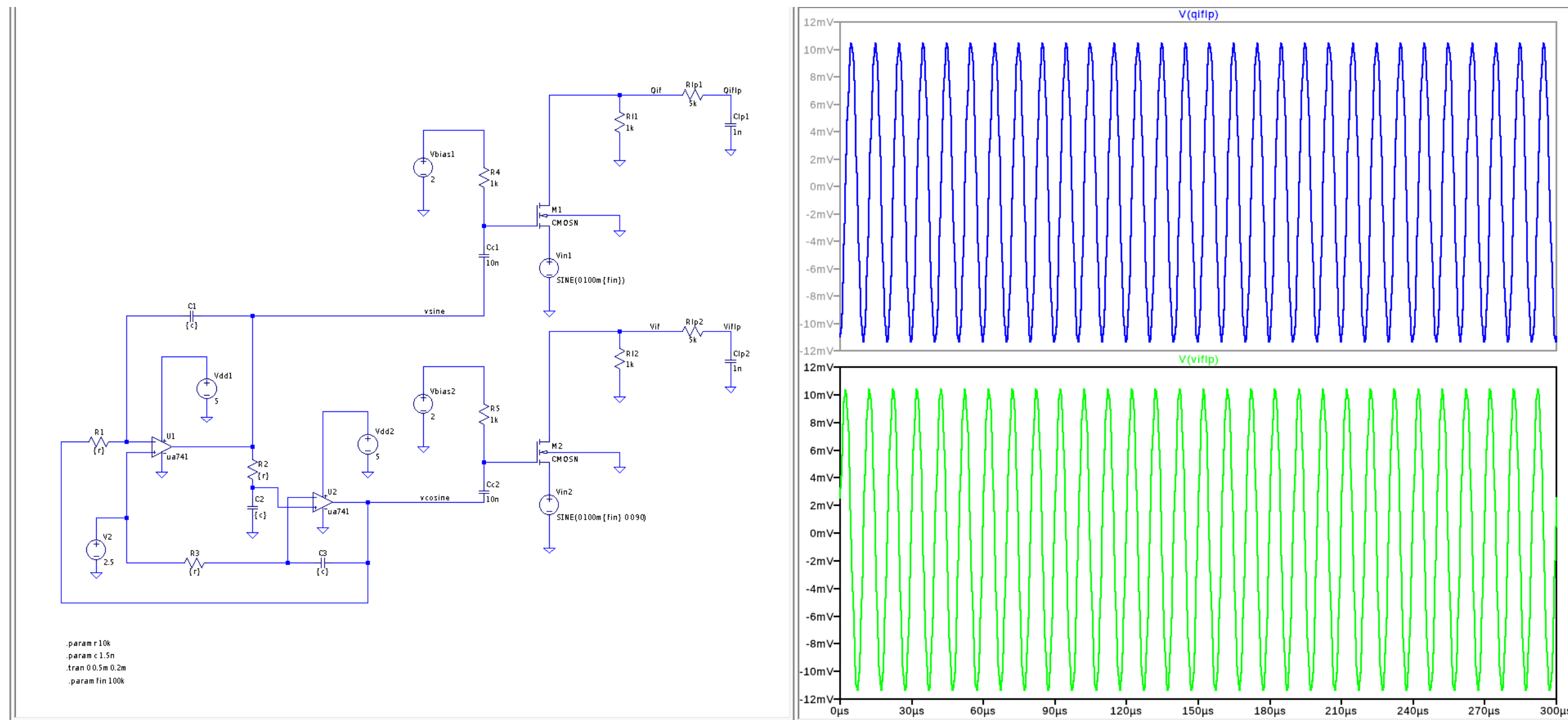
Low Pass Filter



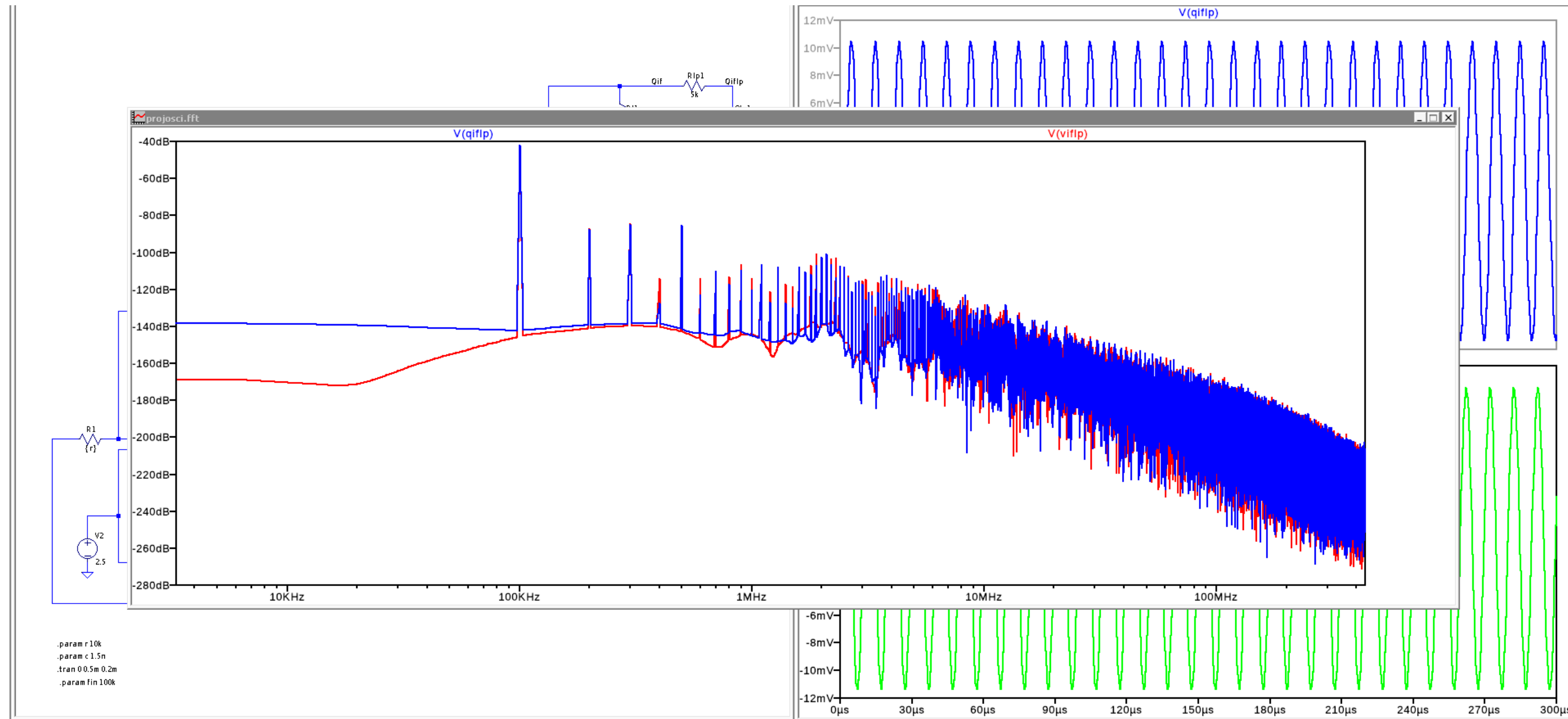
Connecting a low pass filter to the end of the circuit with a cut-off frequency 30 kOhm



Using the filter to filter out higher junk frequencies and keeping the difference frequency



Combining all modules, we get an output that looks similar to two sinusoids



But on frequency analysis, we don't get the difference frequencies and no filtering due to some error